

Basics 24: Derivation of Ideal Magnetohydrodynamics

IV - Magnetohydrodynamics

From multifluid plasma dynamics and Maxwell equations to the closed ideal-MHD system

Textbook draft, version 1.0. Ideal MHD is a controlled long-wavelength, low-frequency reduction of plasma dynamics. This chapter derives its equations, records every major approximation, and emphasizes exactly which kinetic and nonideal effects are removed. This chapter is designed for a reader with no prior plasma-physics training. It distinguishes exact identities from physical approximations and connects the central equations to later simulation modules.

Contents

1	Learning objectives	2
2	Prerequisites and reading strategy	2
3	Target ideal-MHD system	2
4	Mass conservation	3
5	Momentum and magnetic stress	3
6	Ideal Ohm's law as an ordering	3
7	Induction equation	4
8	Thermodynamic closure	4
9	Total energy conservation	4
10	Assumption ledger and domain of validity	5
11	Computational laboratory	5
12	Summary	5

Symbols and acronyms used in this document	6
13 Exercises	6
14 References	6

1 Learning objectives

After completing this note, the reader should be able to:

- State the closed ideal-MHD equations.
- Derive mass and momentum conservation from species equations.
- Reduce generalized Ohm's law to the ideal condition.
- Derive the induction equation.
- Derive pressure and total-energy closures.
- Construct an explicit assumption ledger for model validity.

2 Prerequisites and reading strategy

- Basics 12, 21, 22, and 23; Basics 3 and 5 for vector/PDE manipulations.

On a first reading, emphasize physical meaning and dimensional checks. On a second reading, reproduce the derivations. Attempt the computational laboratory only after the limiting cases are understood.

3 Target ideal-MHD system

A common conservative ideal-MHD system is

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0, \quad (3.1)$$

$$\partial_t (\rho \mathbf{u}) + \nabla \cdot \left[\rho \mathbf{u} \mathbf{u} + \left(p + \frac{B^2}{2\mu_0} \right) \mathbf{I} - \frac{\mathbf{B} \mathbf{B}}{\mu_0} \right] = 0, \quad (3.2)$$

$$\partial_t \mathbf{B} - \nabla \times (\mathbf{u} \times \mathbf{B}) = 0, \quad (3.3)$$

$$\partial_t E + \nabla \cdot \left[\left(E + p + \frac{B^2}{2\mu_0} \right) \mathbf{u} - \frac{\mathbf{B}(\mathbf{u} \cdot \mathbf{B})}{\mu_0} \right] = 0, \quad (3.4)$$

$$\nabla \cdot \mathbf{B} = 0, \quad (3.5)$$

with

$$E = \frac{\rho u^2}{2} + \frac{p}{\gamma - 1} + \frac{B^2}{2\mu_0}. \quad (3.6)$$

We now identify where each equation comes from.

4 Mass conservation

Summing species number equations weighted by mass gives

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (4.1)$$

when nuclear reactions, external injection, and mass loss are absent. This step requires no quasineutral approximation; it follows from total mass accounting.

5 Momentum and magnetic stress

Summing species momentum gives, under quasineutrality and negligible relative-drift inertia,

$$\rho \frac{D\mathbf{u}}{Dt} = -\nabla p + \mathbf{J} \times \mathbf{B}. \quad (5.1)$$

At low frequency, Ampere's law gives $\mathbf{J} = \nabla \times \mathbf{B} / \mu_0$. Using

$$(\nabla \times \mathbf{B}) \times \mathbf{B} = -\nabla \left(\frac{B^2}{2} \right) + (\mathbf{B} \cdot \nabla) \mathbf{B} \quad (5.2)$$

for $\nabla \cdot \mathbf{B} = 0$,

$$\mathbf{J} \times \mathbf{B} = -\nabla \left(\frac{B^2}{2\mu_0} \right) + \frac{(\mathbf{B} \cdot \nabla) \mathbf{B}}{\mu_0}. \quad (5.3)$$

These terms are magnetic pressure and magnetic tension. Rewriting them as the divergence of Maxwell stress yields the conservative momentum equation.

6 Ideal Ohm's law as an ordering

Generalized Ohm's law contains

$$\mathbf{E} + \mathbf{u} \times \mathbf{B} = \eta \mathbf{J} + \frac{\mathbf{J} \times \mathbf{B}}{en_e} - \frac{\nabla \cdot \mathbf{P}_e}{en_e} - \frac{m_e}{e} \frac{D_e \mathbf{u}_e}{Dt} + \dots \quad (6.1)$$

Ideal MHD assumes all right-hand terms are small on the scales of interest:

$$\boxed{\mathbf{E} + \mathbf{u} \times \mathbf{B} = 0}. \quad (6.2)$$

This is not a universal law. It fails at sufficiently small scales, high frequencies, low collisionality, reconnection layers, and kinetic resonances.

7 Induction equation

Insert ideal Ohm's law into Faraday's law:

$$\frac{\partial \mathbf{B}}{\partial t} = -\nabla \times \mathbf{E} = \nabla \times (\mathbf{u} \times \mathbf{B}). \quad (7.1)$$

Together with $\nabla \cdot \mathbf{B} = 0$, this transports and stretches magnetic flux with the fluid. Expanding the curl gives

$$\frac{D\mathbf{B}}{Dt} = (\mathbf{B} \cdot \nabla)\mathbf{u} - \mathbf{B}(\nabla \cdot \mathbf{u}), \quad (7.2)$$

after using mass conservation and zero magnetic divergence in the appropriate form.

8 Thermodynamic closure

For an adiabatic ideal gas with no heat conduction or dissipation,

$$\frac{Dp}{Dt} + \gamma p \nabla \cdot \mathbf{u} = 0, \quad (8.1)$$

equivalently

$$\frac{D}{Dt} \left(\frac{p}{\rho^\gamma} \right) = 0. \quad (8.2)$$

The parameter γ represents the chosen closure, commonly 5/3 for an isotropic monatomic gas. Collisionless magnetized plasmas may not justify this scalar adiabatic law.

9 Total energy conservation

Dot the momentum equation with $\mathbf{u}$, combine the internal-energy equation with Poynting's theorem, and use ideal Ohm's law. The result is

$$\partial_t E + \nabla \cdot \left[\left(E + p + \frac{B^2}{2\mu_0} \right) \mathbf{u} - \frac{\mathbf{B}(\mathbf{u} \cdot \mathbf{B})}{\mu_0} \right] = 0. \quad (9.1)$$

The flux contains advected kinetic/internal energy, pressure work, and electromagnetic transport. A numerical solver that updates the separate pieces inconsistently can violate this balance.

10 Assumption ledger and domain of validity

Ideal MHD commonly assumes:

1. quasineutrality and negligible displacement current;
2. scales large relative to Debye length and relevant gyro radii;
3. frequencies below electron and often ion kinetic scales;
4. sufficiently small Hall, resistive, electron-inertia, and anisotropic-pressure effects;
5. a scalar thermodynamic closure;
6. a continuum description with enough particles per resolved volume.

The present project uses ideal MHD for bulk Alfvén-wave structure, then adds energetic-particle kinetics because resonant fast-ion drive lies outside this closure.

11 Computational laboratory

Create an assumption-ledger notebook that evaluates dimensionless ratios for a selected fusion-plasma case. Numerically verify the magnetic-force identity and conservative stress form on a smooth divergence-free field. Integrate a one-dimensional ideal-MHD wave benchmark and monitor total energy.

12 Summary

- Ideal MHD is derived by combining species moments, low-frequency Maxwell equations, and an ordered Ohm law.
- Magnetic pressure and tension arise from the Lorentz force.
- Ideal Ohm’s law produces the induction equation and frozen magnetic flux.
- A scalar adiabatic law closes the thermodynamics.
- Every ideal-MHD result inherits a clear scale and closure domain.

Symbols and acronyms used in this document

Symbol / acronym	Name	Meaning in this document
$\rho, \mathbf{u}, p$	fluid variables	Mass density, bulk velocity, and scalar pressure.
E	total energy density	Kinetic, internal, and magnetic energy.
γ	adiabatic index	Thermodynamic closure parameter.
η	resistivity	Nonideal coefficient omitted in ideal MHD.
D/Dt	material derivative	$\partial_t + \mathbf{u} \cdot \nabla$.
$\mathbf{I}$	identity tensor	Unit second-order tensor.

13 Exercises

1. Derive the magnetic-pressure and tension decomposition of $\mathbf{J} \times \mathbf{B}$.
2. Derive the induction equation from Faraday and ideal Ohm laws.
3. Show that p/ρ^γ is advected for the adiabatic closure.
4. List the terms omitted from generalized Ohm's law and associate each with a physical scale.
5. Explain why ideal MHD can describe an Alfvén eigenfunction but not energetic-particle resonant drive by itself.

14 References

References

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